

Developing a TEWL-Measuring Device and Using Skin Mold Imaging to Quantify TEWL

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Background and Motivation

Transdermal drug delivery (TDD) is an non-invasive drug delivery method achieved by applying a patch, cream, or gel to the skin to deliver medication into a patient's bloodstream. TDD is dependent on the drug's ability to permeate through the stratum corneum (SC)—the outermost layer of the skin that functions as the skin's barrier against toxins and pathogens.

The permeability of the SC is often characterized by the concept of transepidermal water loss (TEWL), which is a measure of the passive diffusion of water vapor from the SC. TEWL is an indicator of how intact the SC is, providing insight into the skin's ability to retain moisture and uphold barrier function. TEWL is typically measured with TEWL devices, which do so by assessing time-dependent water evaporation. However, these devices are often expensive, typically costing between \$8000 and \$12000. Thus, numerous research labs have been prompted to develop their own TEWL measurement devices instead, including our own lab.

TEWL-measuring devices cannot be used in isolation as the real skin surface area from which TEWL is being measured is unknown due to skin furrows. The SC contains non-uniform, microscopic furrows, and stretching the skin can flatten out these furrows and widen the gaps between corneocytes. To determine the real surface area, the procedure involves placing ExaFlex molds on stretched and unstretched skin, followed by imaging the molds with a confocal laser scanning microscope and processing it with ImageJ to obtain 3D models and the surface area values.

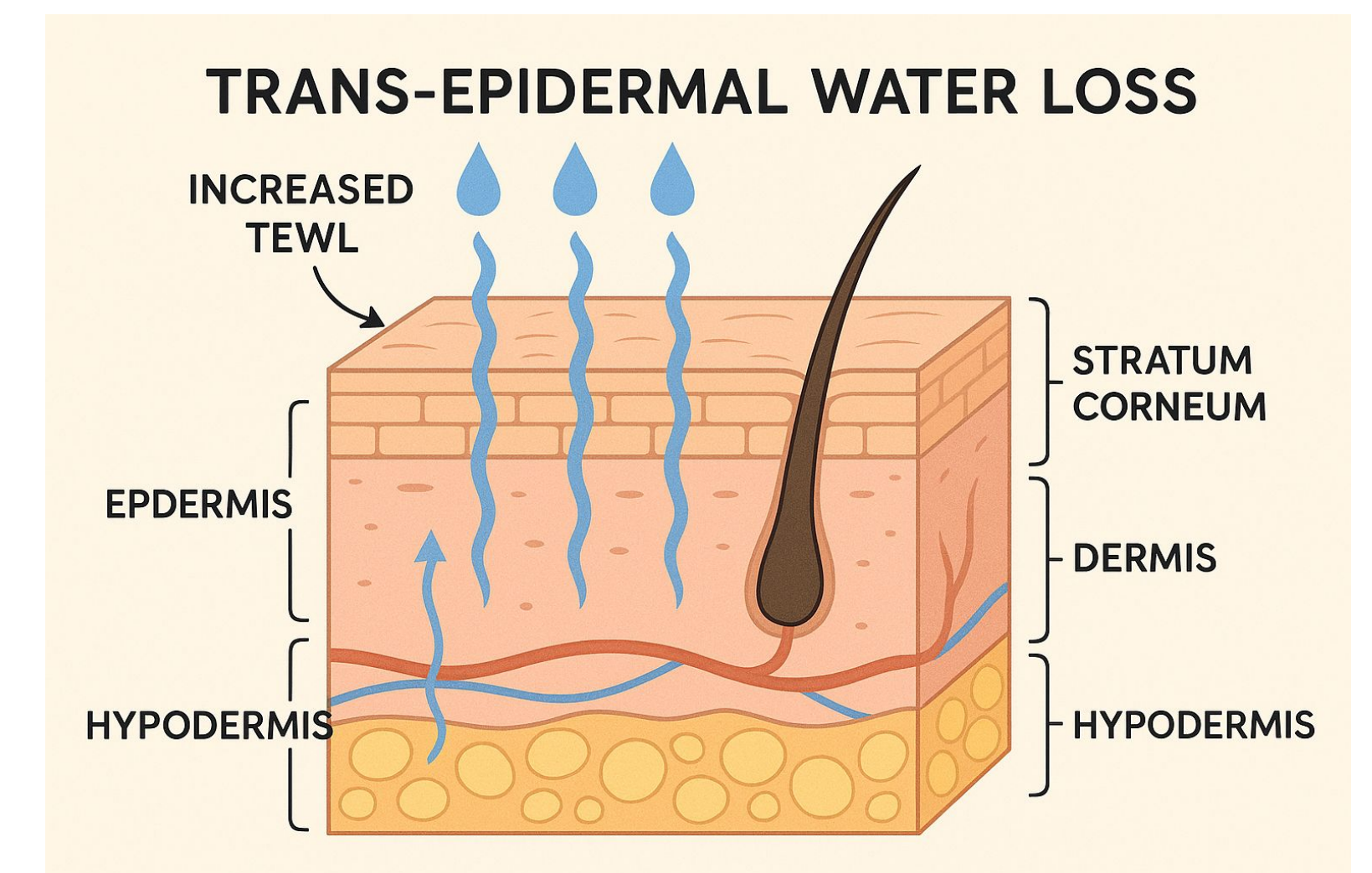


Figure A: Diagram of TEWL and the structure of human skin

TEWL Device Methods

Our lab's retail TEWL-measuring devices—the Vapometer and Aquaflux—have been incapable of recording accurate TEWL measurements consistently, despite over a year of testing. Thus, our lab has been referencing the article WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss as a foundation to build a TEWL measuring device [6]. It operates by placing a closed resin chamber on the subjects arm. The micropump pumps air into the chamber flushes skin residuals until the RH sensor reads %. The RH sensor and thermocouple begins to collecting data for 60 seconds. Data is processed in MATLAB where RH values are converted into TEWL values using a calibration curve. Figure C illustrates the schematic of the TEWL-measuring device circuitry.

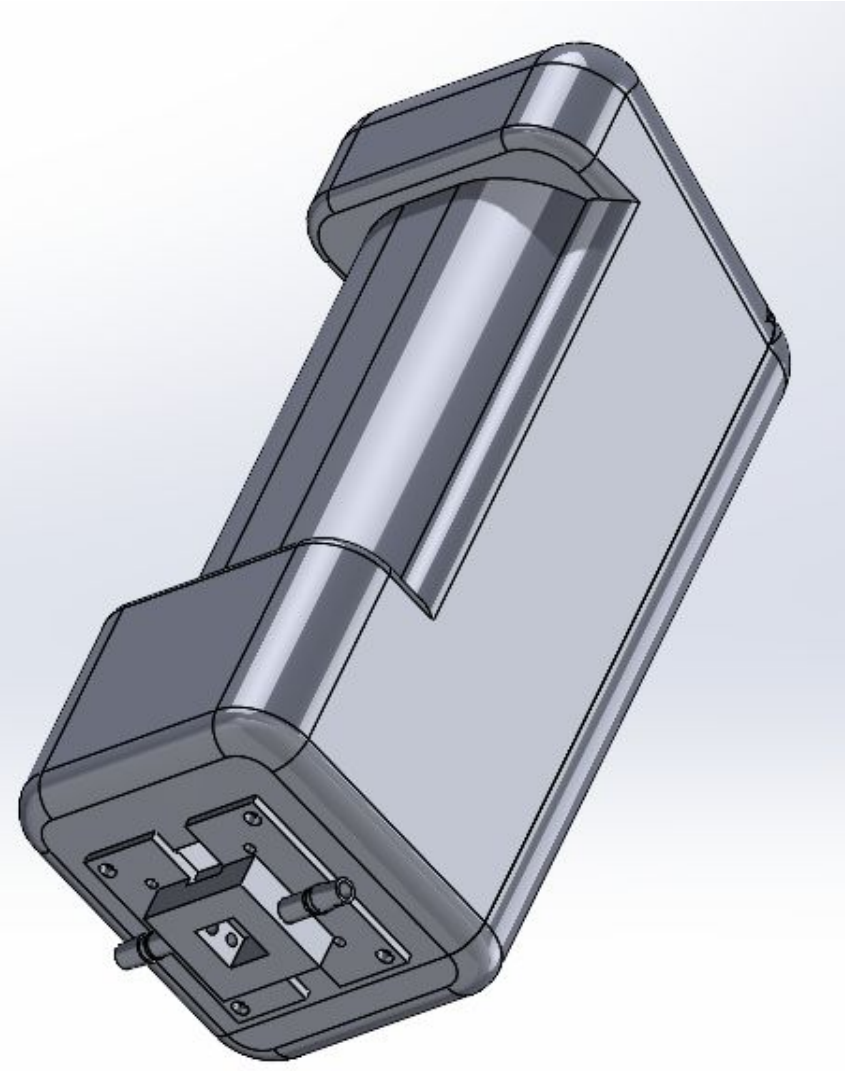


Figure B. 3D-printed TEWL device prototype

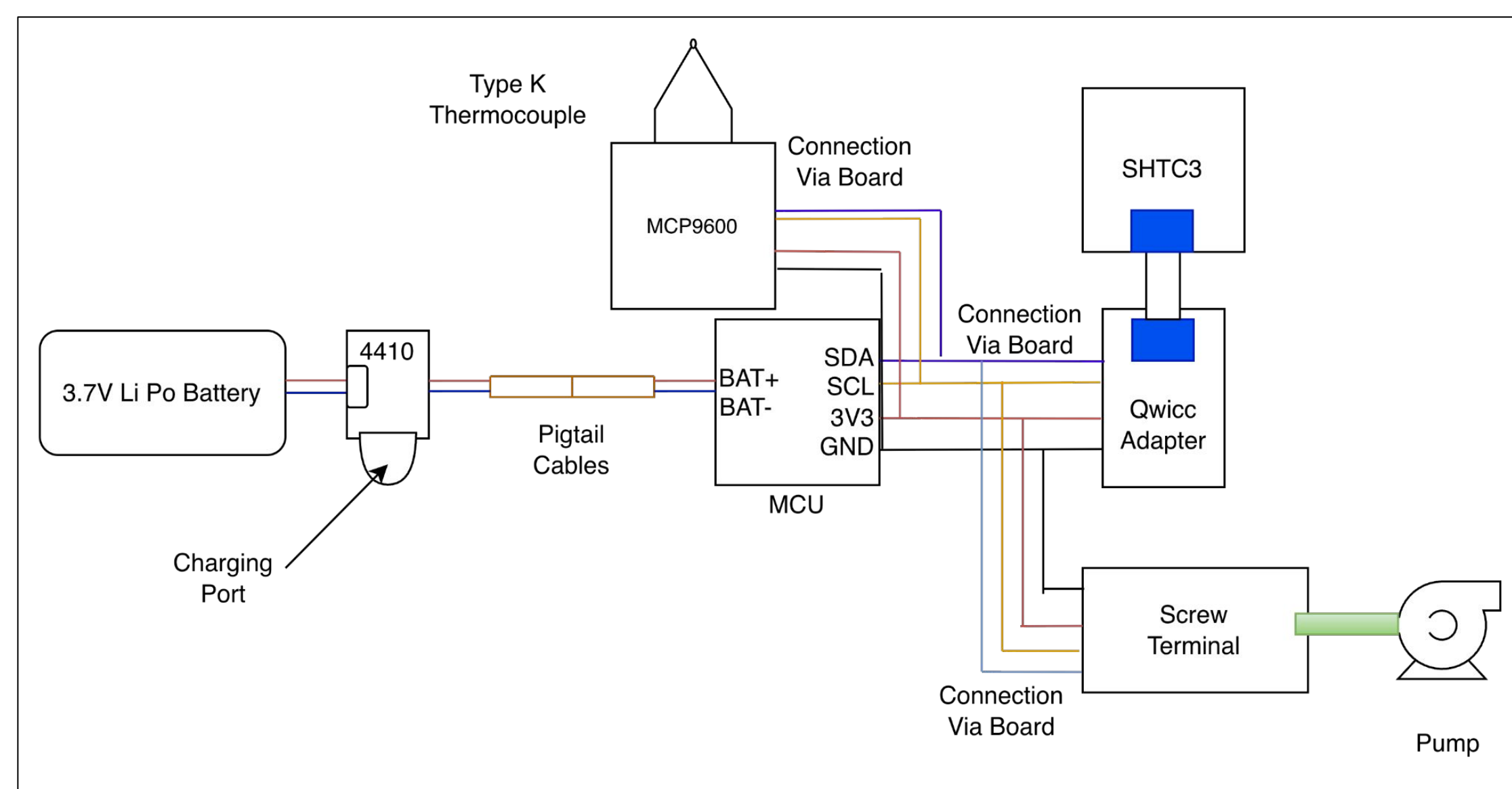


Figure C. Schematic of the TEWL device's circuitry.

Skin Imaging Methods

To perform skin mold imaging, skin molds are first created by placing ExaFlex molds onto a test participant's dorsal forearm within a 3D-printed skin mold template (Figure D). The skin mold is then glued onto a petri dish filled with Milli-Q water and imaged with the Zeiss LSM510 Confocal Laser Scanning Microscope (CLSM), which takes images at different heights to create a Z-stack. This Z-stack is then processed to remove noise in the ImageJ software, where the BoneJ plugin then computes the surface area of the skin.



Figure D: Skin mold being created on a test participant using the 3D-printed template

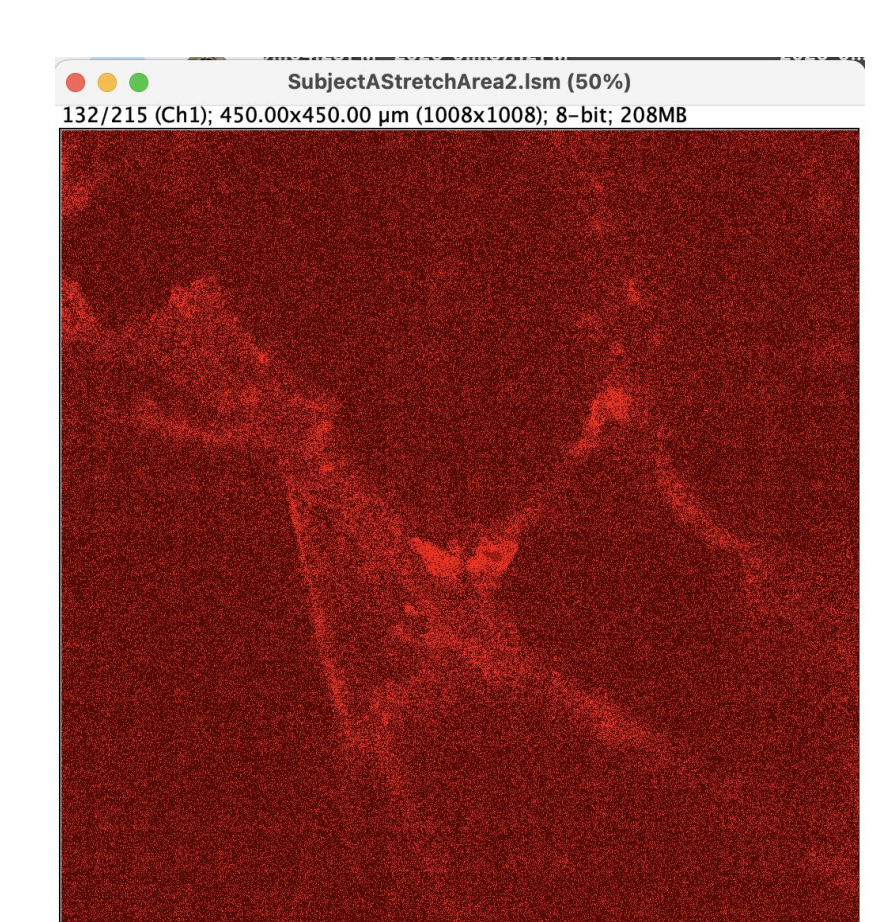


Figure E: One image on an example Z-stack created using the CLSM

Skin Imaging Discussion

The lab used the control skin mold (see Figure D) to modify the image analysis procedure to reduce noise in the image. The process involved reducing the brightness/contrast from 0 to 1 as the LUT histogram showed only two colors present on the scale. Additional added image processing included applying the *smooth* and *sharpen* feature to blur the active image and to take the weighted average of neighboring pixels respectively to increase the images contrast. From here, binarization using the IJ ISO Data setting followed by the Median filter was applied to the image. As a result of these modifications, the processed image had undergone substantial noise reduce as shown in Figure F and G.

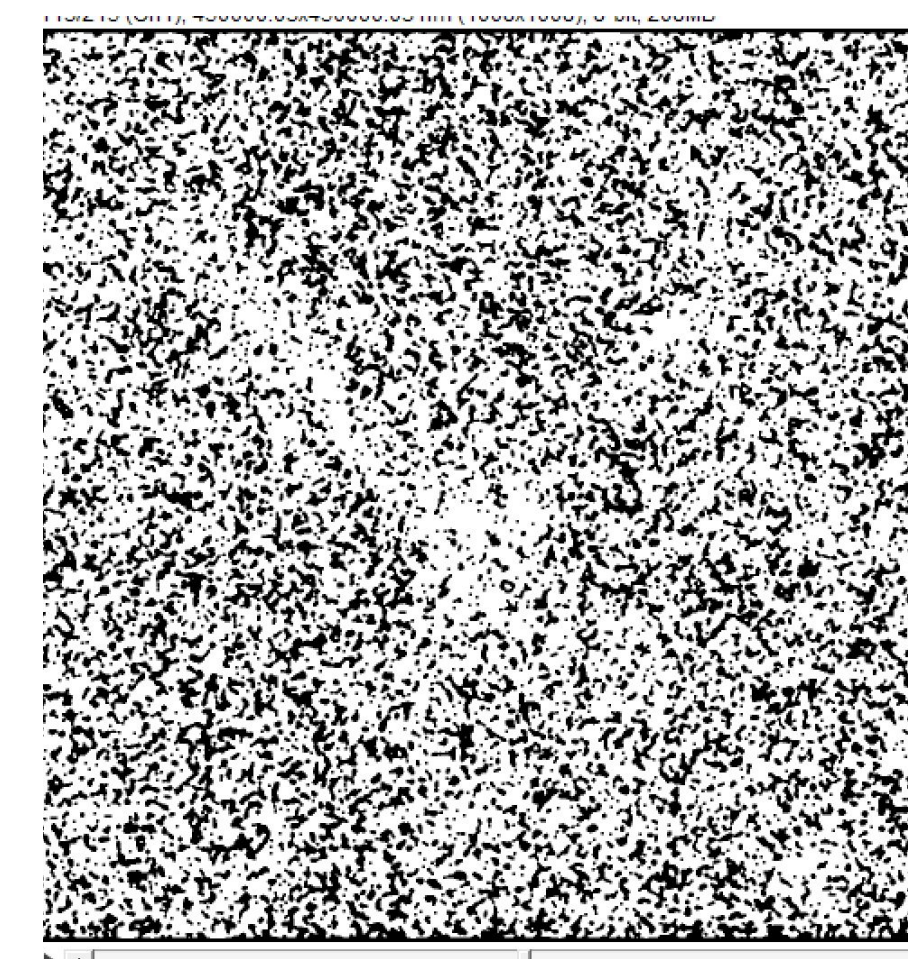


Figure F: Image using the previous image processing procedure

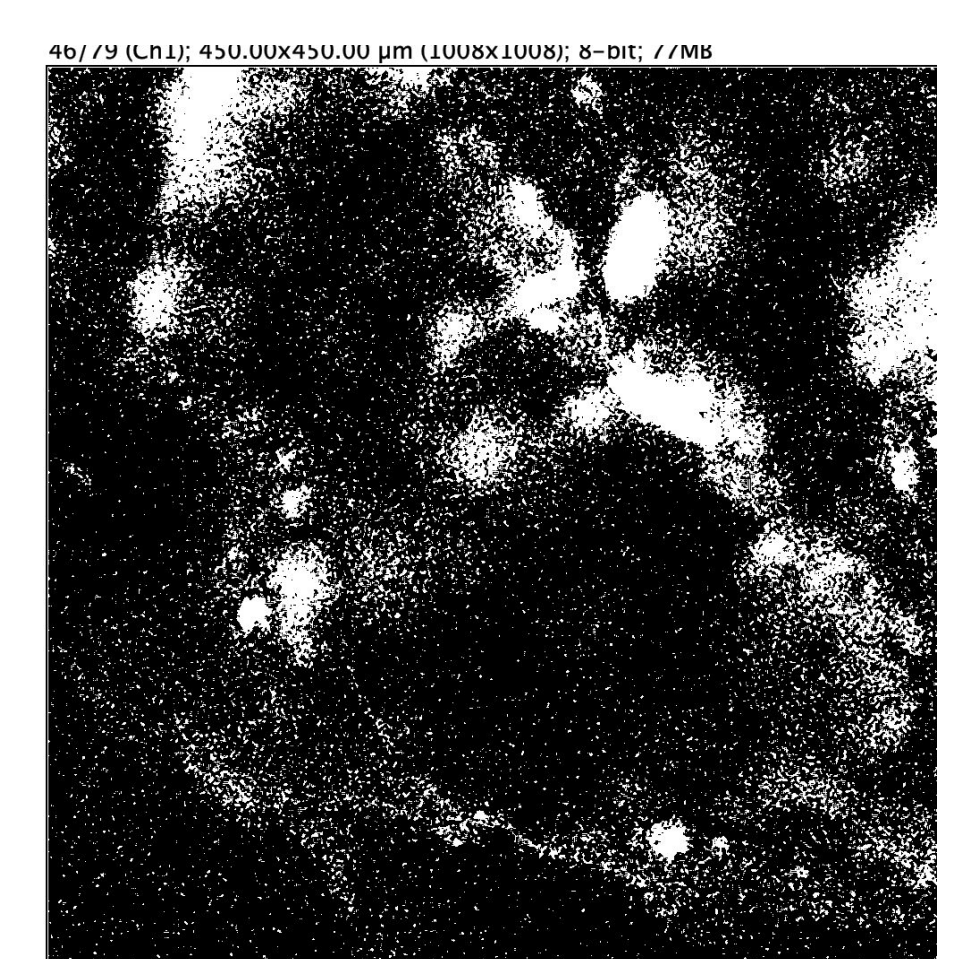


Figure G: Image using the new image processing procedure

TEWL Device Results

It was found that the micropump was not strong enough to flush out the moisture accumulated on the relative humidity RH sensor after placing the 3D-printed chamber on skin (as seen in Figure H). While lab experimented with a hand pump and hair dryer, it was not sufficient at flushing out the moisture.

The RH sensor also took over 1000 seconds to return back to the baseline RH after being placed on the arm. This prompted the efficacy of the RH sensor to be questioned, as the manual for the sensor stated that steady state should be achieved after 20 to 30 seconds after being placed in standard conditions.

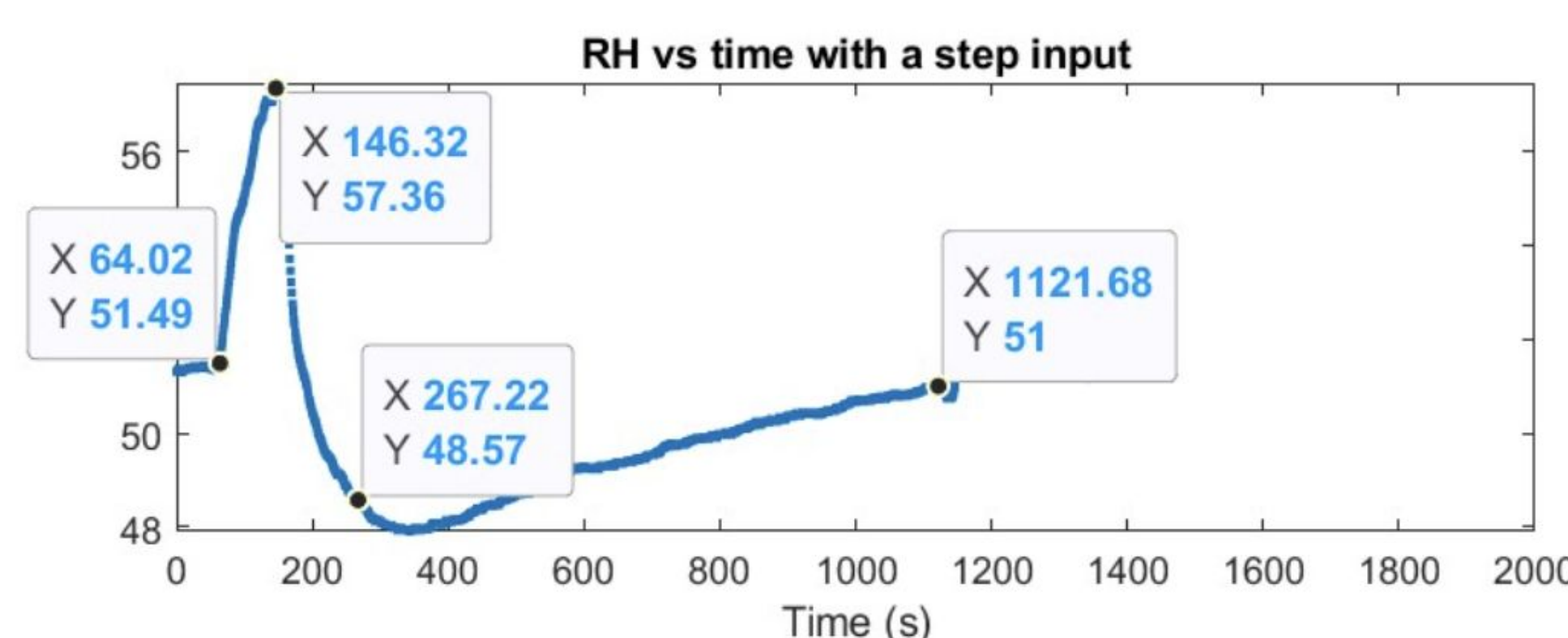


Figure H: RH vs. time

Future Work

The micro pump does not flush out any skin residuals as there is close to 0 air flow coming out of the pump. The lab is looking for alternative micropumps and air drying strategies. Additionally, the relative humidity sensor does not operate when it is placed in the resin chamber designed for the TEWL measuring device. Given that it does still work in an open air setting, the lab will be redesigning the chamber to ensure that the relative humidity sensor can operate within the chamber setting. The lab will then obtain a calibration curve using the wet cup method and begin user testing.

The skin imaging procedure requires further modifications as it is still incomplete and fails to output consistent surface area readings. Future work would involve identifying the pixel sizes that corresponds to the lengths of the skin molds. Additionally, the lab would work on finding the optimal 3D model that visualizes the furrow patterns on the subjects skin.

References and Acknowledgements

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